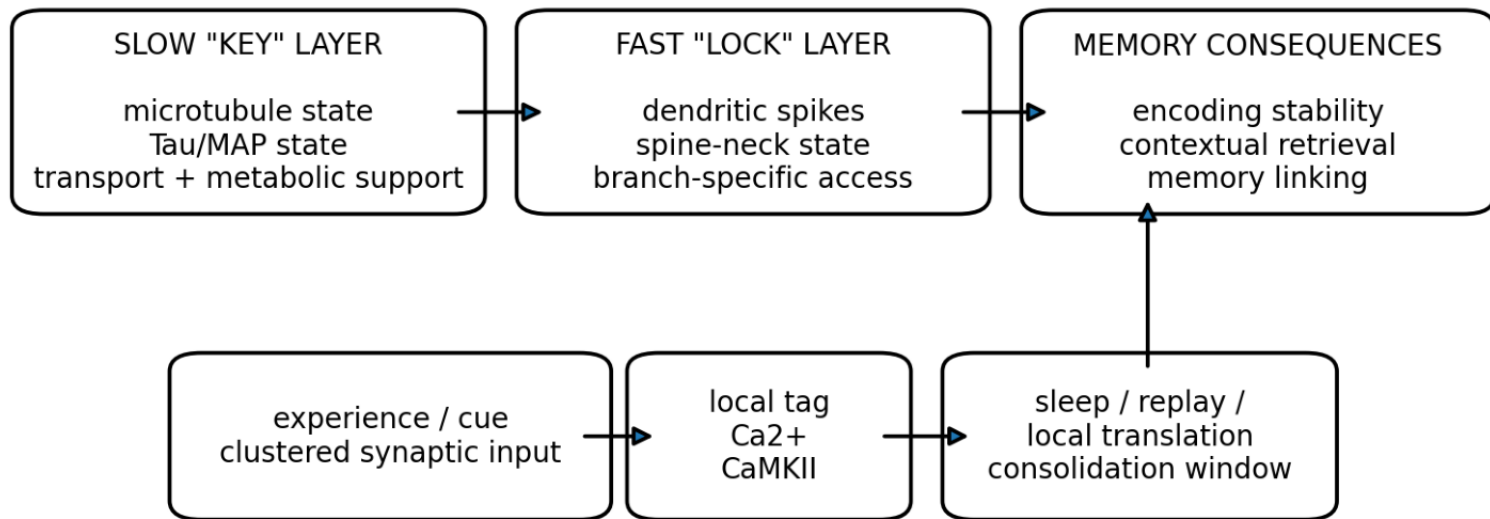


Cytoskeletal-Dendritic Key-Lock Model of Associative Memory

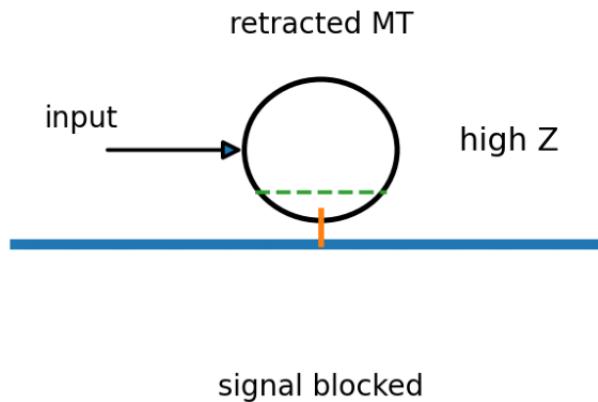
Slow structural accessibility beneath fast dendritic dynamics



Core claim: memory access depends not only on synaptic strength,
but also on a slowly learned branch-level accessibility structure

Figure 1. Structural impedance gating

A. Mismatched key



B. Matched key

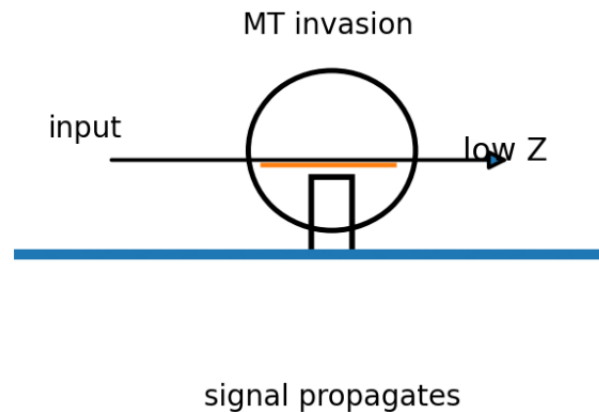
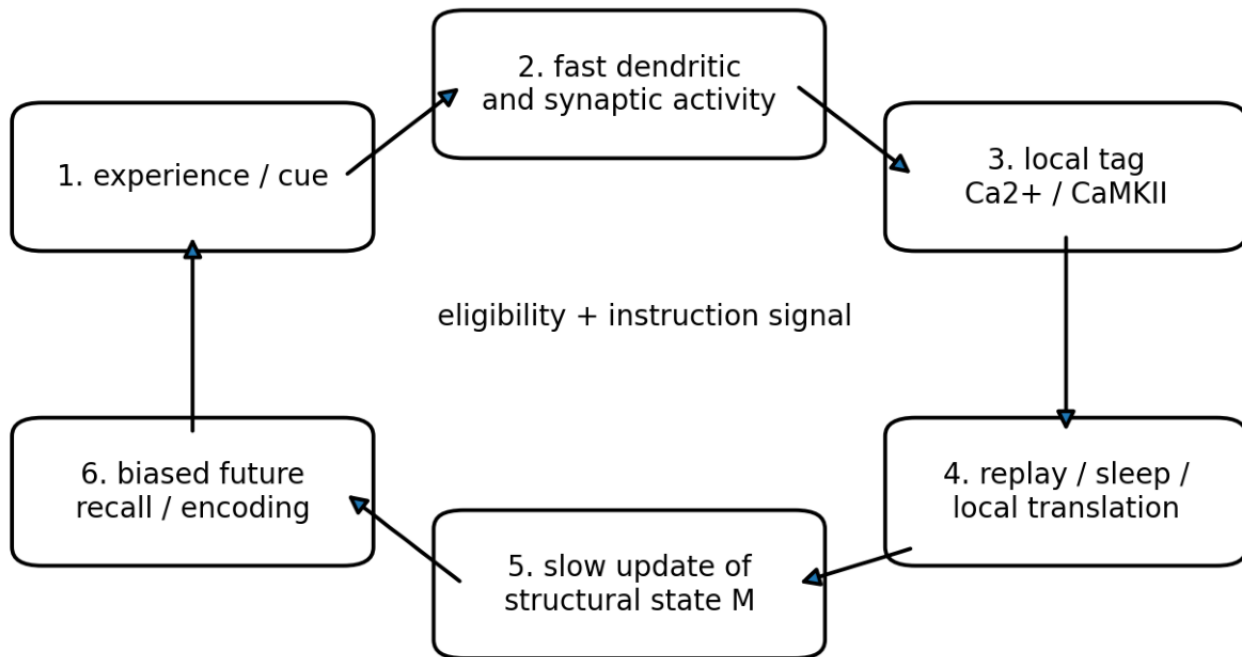
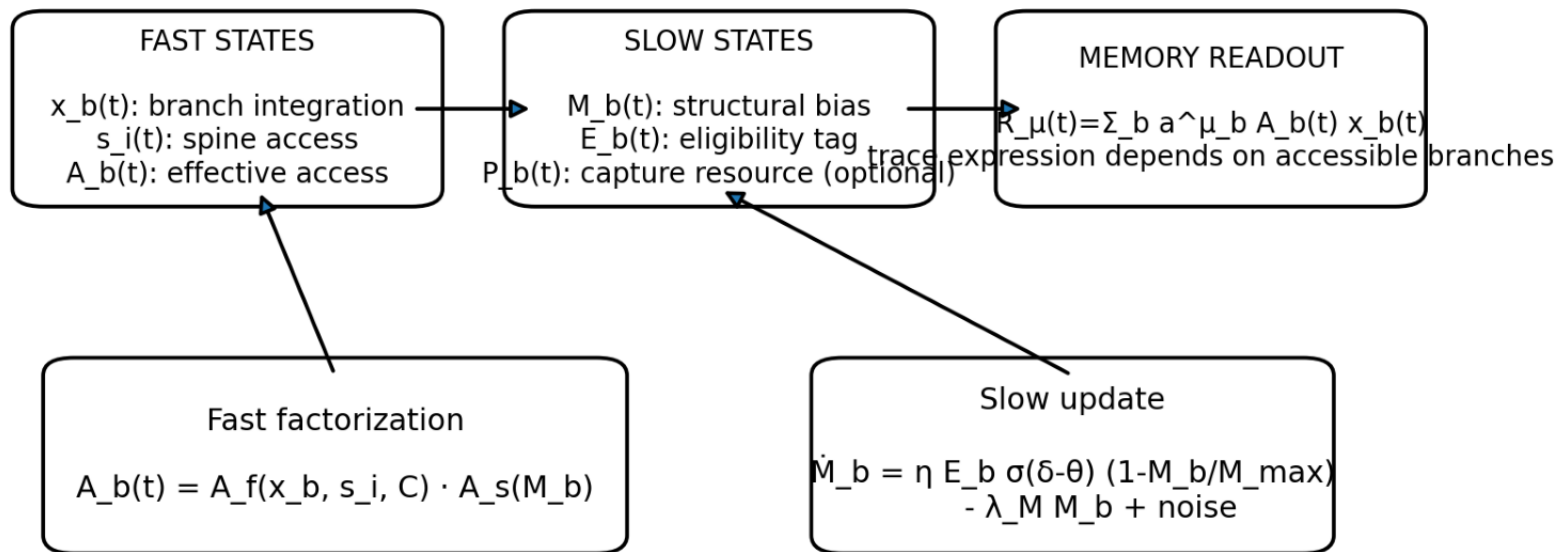


Figure 2. Cytoskeletal learning cycle



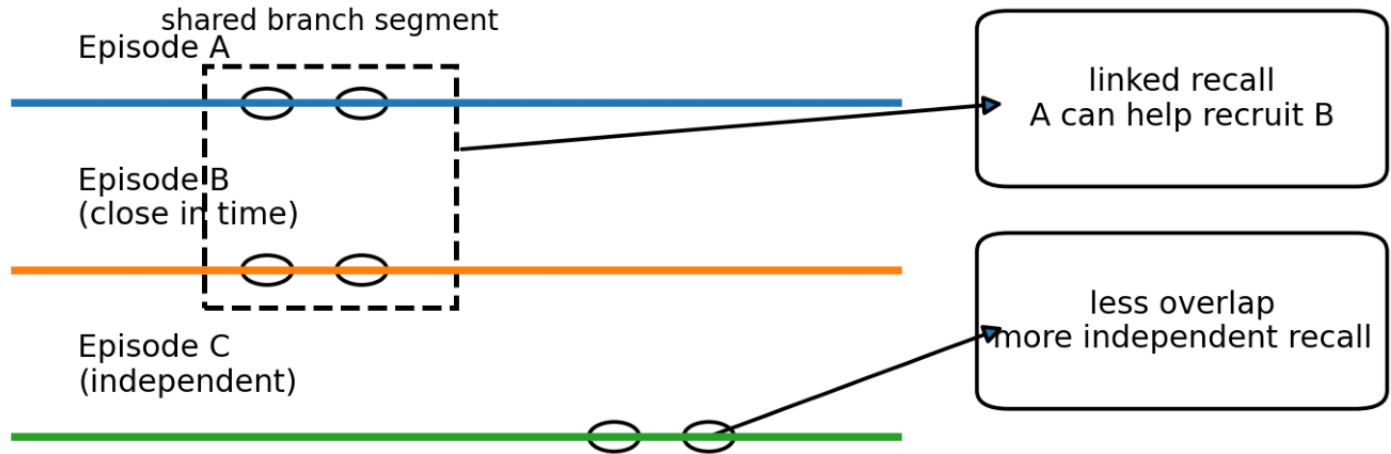
plastic changes accumulate slowly, while access opens fast

Figure 3. Minimal biological formalization



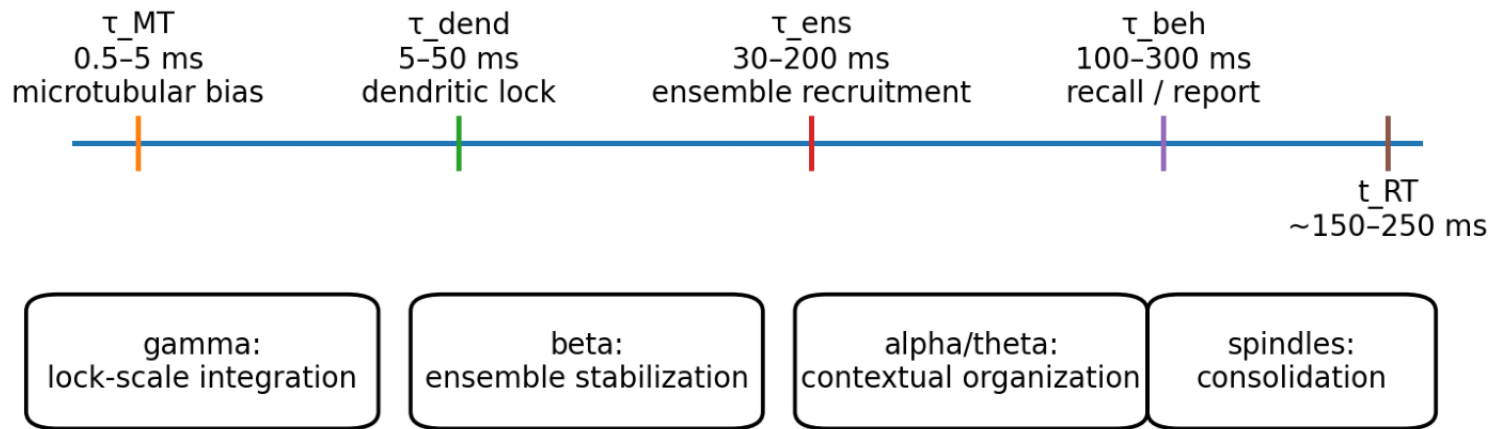
Interpretation: synapses do not disappear; rather, the relevant branch subset can become less accessible.

Figure 4. Memory linking and contextual retrieval



Close memories can reuse accessible dendritic segments; distant or unrelated memories need not.

Figure 5. Temporal cascade from structural bias to behavioral recall



The model claims a nested timing hierarchy, not an instantaneous one-step lookup.